

VI!TM



Voice Streaming



Voice Streaming

What is Streaming?

- Streaming is the process of sending audio data in small continuous chunks instead of waiting for the entire file to download. In our case, we are using WebSockets for streaming.
- Types of Streams:
 - Bidirectional Stream
 - Audio flows both ways (VI ↔ Endpoint)
 - Use cases: AI voice bots, conversational IVR
 - Unidirectional Stream
 - Audio flows one way (VI → Endpoint)
 - Use cases: live transcription, real-time monitoring, coaching

Why it matters for voice calls:

- Enables real-time experiences (like hearing someone as they speak).
- Reduces delay (latency), making conversations smoother.
- Supports use cases like live conversational AI, transcription, call monitoring and coaching..



Web Socket

WebSocket is a communication protocol that allows persistent, full-duplex (two-way) communication between a client and server over a single connection.

Why it's used in streaming:

- Keeps the connection open and active throughout the call.
- Allows instant push and pull of audio packets in real-time.
- Works much better than traditional APIs for continuous data flow.

Normal API (HTTP):	WebSocket:
• Works on a request → response model. • Each call opens and closes a new connection. • Higher latency, not suitable for live audio.	• Creates a single long-lived connection. • Server and client can both send data at any time. • Designed for real-time, low-latency communication.

Chunk Size in Streaming

What is Chunk Size?

- A chunk = a small packet of audio data sent during streaming.
- Instead of sending the whole audio at once, we break it into continuous chunks for real-time delivery.

Conditions for Chunk Size

- **Minimum:** 1.6 KB (~100 ms audio)
 - Smaller chunks may cause jitter or unstable audio.
- **Maximum:** 50 KB
 - Larger chunks may lead to delays or connection timeouts.
- Must be a **multiple** of 160 bytes
 - Each 160 bytes = 10 ms of audio, ensuring proper alignment.

Always keep chunk size between 1.6 KB and 50 KB, in multiples of 160 bytes — this guarantees smooth, real-time streaming.

Streaming Protocol

1. Communication Channel

- Voice streaming between VI and customer endpoint happens over a WebSocket connection.
- Supports both:
 - **ws://** - non-secure
 - **wss://** - secure (encrypted)

2. WebSocket URL Configuration

- Static URL: A fixed endpoint, e.g., **wss://yourservice.com/stream**
- Dynamic URL: Can be set using variables (fetched via API from customer systems).

3. Custom Parameters

- Up to 3 key-value pairs can be added.
- These are passed in the start event body when streaming begins.
- Useful for passing identifiers like: Unique ID, Agent ID, Product etc,

4. Message Format : All messages are exchanged as JSON strings.

Streaming Configuration - Bidirectional

- Enabled via **Streaming Object** (only).
- Always runs in **Foreground Mode** (cannot be Background).
- Stream ends when:
 - The call disconnects,
 - WebSocket closes,
 - Client explicitly stops it, or
 - “Total Stream Duration” completes.

Streaming Configuration - Unidirectional

→ Using Streaming Object

- **Foreground Mode**

- The call flow pauses while the stream is active.
- Stream ends when:
 - ◆ The call disconnects,
 - ◆ WebSocket closes,
 - ◆ Client stops it, or
 - ◆ “Total Stream Duration” is reached.

- **Background Mode**

- The stream runs in the background.
- The call flow continues to the next object while streaming happens simultaneously.

→ Using API

- Incoming calls
- Outgoing calls
- Call patching
- Click-to-Call scenarios

Media Format Requirements

Parameter	Details
Format	WAV file. Raw PCM (Pulse Code Modulation) – uncompressed audio for maximum quality and low latency.
Encoding	16-bit Linear PCM – each audio sample is represented by 16 bits (2 bytes).
Sample Rate	8 kHz – 8,000 samples per second, standard for telephony audio.
Bitrate	128 Kbps – fixed data rate ensuring smooth playback.
Encoding Method	Base64 – audio is encoded into Base64 strings for safe transfer inside JSON messages.
Direction	Both inbound and outbound must use the same format (caller → VI and VI → caller).
Compatibility	Ensures real-time, lossless audio exchange between VI and customer endpoint.



WebSocket Messages (VI → Endpoint)

Event	Purpose
connected	Sent when WebSocket is established. Confirms session has started.
start	Sent after <i>connected</i> . Includes call ID, room ID, caller info, media format, and custom parameters.
media	Carries audio in Base64-encoded PCM packets. Includes chunk index & timestamp.
dtmf (<i>Bidirectional only</i>)	Sent when caller presses a keypad digit. Used in IVR/voice bot flows.
stop	Indicates the stream has ended (call disconnected, WebSocket closed, client stopped, or duration elapsed).
mark (<i>Bidirectional only</i>)	Checkpoint event confirming audio playback completion.



WebSocket Messages (Endpoint → Vi)

Event	Purpose
media	Sends audio back to VI in Base64-encoded PCM format. Must match the same format as inbound audio.
mark	Requests acknowledgment once certain audio has finished playing.
clear	Clears any queued audio that has not yet been played out.
exit	Terminates the active WebSocket session in an orderly way, allowing IVR logic to continue.

Thank You

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